



Standard Operating Procedure

OH-234 Determination of Dietary Ingredient Overages for Supplements and Natural Health Products

Document Owner:		Brian Laster
Owning Function/Organization:		R&D, Process & Formulation, MPD
Applicable Business: ☐ Oral Care (Brush) ☐ Oral Care (Formulated) ☒ Personal Health Care Shareable with EBPs? ☒ Yes ☐ No	Applicable Region: ☐ Asia Middle East Africa ☐ Europe ☐ Greater China ☐ Latin America ☒ North America	Applicable Product Category: ☐ Cosmetics ☒ Dietary Supplements ☐ Food Supplements ☐ Household Products ☐ Medical Devices ☐ Medicinal Products/Drugs ☒ Natural Health Products ☐ Other (see scope)

PURPOSE

For the United States, general provisions for Food Labeling are defined in 21 CFR 101, *Food Labeling*. According to the Dietary Supplement Health and Education Act (DSHEA), Dietary Supplement products are regulated under the general umbrella of “foods”. Therefore, these General Provisions are in scope for Supplement products. Within 21 CFR 101.9 (g)(4), limits are established for fortified nutrients declared on food product labels to ensure that the food product is not misbranded.

These nutrient types and corresponding limits are:

1. Added nutrients intended for dietary fortification are considered Class I nutrients (i.e., Vitamin C derived from Ascorbic Acid). Class I nutrients must be no less than 100% of the labelled claim throughout the expiration period. Formulated Overages may be applied to Class I nutrients to ensure the regulations outlined in 21 CFR 101 are met.
2. Added nutrients intended for dietary fortification, but naturally occurring (indigenous) to an ingredient (i.e., Vitamin C derived from Acerola Cherry Extract), are considered Class II nutrients. Class II nutrients must be no less than 80% of the labelled claim throughout the expiration period. Formulated Overages may be applied to Class II nutrients to ensure the regulations outlined in 21 CFR 101 are met.
3. Third Group Nutrients with mandatory labelling requirements present in the finished product (i.e., calories), but not intentionally added for fortification do not require formulated overages. Third Group nutrients must be no more than 120% of the labelled claim. Product labelling should reflect known or expected variation of the nutrient in the finished product to ensure labelling requirements are met.

For Canada, there is no distinction between ingredient classes. For Canadian Natural Health Products, section 2.3 of the *Quality of Natural Health Products Guide (QNHPG)* provides nutrient specification guidance when compendia references¹ are not available. The general limits are:

1. Vitamins and Minerals apply a general tolerance limit of 80 – 120 percent of labelled claim.
2. Isolates and Synthetic duplicates (i.e., botanical chemical constituents) apply a general tolerance limit of 80 – 120 percent of labelled claim.
3. Live microorganisms (i.e., probiotics) apply a tolerance limit of 80 percent of the labelled claim (CFU count).
4. Enzymes apply a tolerance limit of 80 – 150 percent of the activity of enzymes labelled claim.



Standard Operating Procedure

OH-234 Determination of Dietary Ingredient Overages for Supplements and Natural Health Products

For Natural Health Products, upper limits above the suggested Canadian upper limit (or compendia limit) may be used if scientifically justified. The scientific basis for these instances is documented per instructions provided in Section 2 of this SOP.

To comply with these requirements, 21 CFR 101.9 (g)(6), 21 CFR 101.36(f)(1), and section 2.3 of QNHPG state that “reasonable excesses” of nutrients are acceptable “within current good manufacturing practice.” The regulations do not define what constitutes “reasonable excesses . . . within current good manufacturing practice” but suggest that excesses (overages) may be applied throughout the duration of product expiry to account for naturally occurring ingredient degradation, manufacturing losses and sources of variation within processing or testing. To comply with good manufacturing practices, any intentional overages must be recorded in manufacturing records. This SOP details the considerations for establishing a dietary ingredient overage during product development and provides instruction on documenting this information in standards, manufacturing records, and testing requirements. A mathematical framework for overage determination is provided that may be utilized when overages need to be established or reviewed.

¹Compendia limits are provided in *OH-229 Product Specifications for Dietary Supplement and Natural Health Products*

SCOPE

This Standard Operating Procedure (SOP) applies to all products classified as Dietary Supplements and Natural Health Products in the North American Personal Health Care business. Products distributed into the United States and Canada are in scope regardless of the region of manufacture or packing.

It applies to:

- Bulk and Finished Goods developed by P&G and produced at P&G manufacturing plants.
- Bulk and Finished Goods developed by P&G and produced at external business partners.
- Bulk and Finished Goods co-developed (Connect & Develop) or sourced (Connect & Go) through external business partners (EBP) when the product composition is not proprietary.

It does NOT apply to:

- Product classifications outside of Dietary Supplement and Natural Health Products, including but not limited to, over the counter drug products, pharmaceuticals, medical devices, food (excluding Dietary Supplement), cosmetics, and consumer goods.
- Regions outside of North America.
- Materials which do not meet the definition of Dietary Ingredients (i.e., other ingredients, excipients).
- Bulk and Finished Goods co-developed (Connect & Develop) or sourced (Connect & Go) through external business partners when the composition is proprietary to the partner. In these instances, the external business partners procedures are followed. P&G oversight is provided through product composition reviews conducted by the Global Product Stewardship organization during market clearance activities and on-going P&G Quality oversight of the EBPs manufacturing and change management.



Standard Operating Procedure

OH-234 Determination of Dietary Ingredient Overages for Supplements and Natural Health Products

RESPONSIBILITIES

PROCESS AND FORMULATION / MATERIALS, PROCESS, DELIVERY (R&D)

- Establishes formulated overages for dietary ingredients.
- Establishes product specifications.
- Documents the basis for the overage as required by Attachment 1.

ANALYTICAL / MICROBIOLOGY

- Determines method suitability for dietary ingredient analysis in finished products.

STABILITY

- Technical oversight of stability studies
- Provides analysis to statistically predict dietary ingredient loss over declared shelf-life.

PRODUCT SAFETY

- Evaluates proposed overage during product development. Documents assessment in the market clearance prior to commercial distribution.

REGULATORY

- Provides guidance on applicable regulations based on dietary ingredients, label claims, and region of sale.

INITIATIVE/CATEGORY QUALITY ASSURANCE

- Approval of Basis for Dietary Ingredient Overage assessments

PROCEDURE

1. P&G determines overages for dietary ingredients based on the anticipated loss of dietary ingredients due to degradation during the shelf life and processing, as well as inherent variance in the manufacturing process and analytical or microbiological methods. Variances are defined as any attribute that may cause the dietary ingredient value to deviate from the intended formulated target.
 - 1.1. Anticipated loss of dietary ingredient(s) due to naturally occurring ingredient degradation (ΔX) is determined by evaluating product stability data. All stability data types including related technical assessments, scientific literature, developmental stability, confirmatory stability, and on-going stability may be leveraged to guide the determination of anticipated loss over shelf-life.
 - 1.2. Anticipated loss of dietary ingredient(s) due to known processing losses (ΔY) are determined by quantifying losses which occur during product manufacturing processes. This loss is determined by assessing batch manufacturing data at pilot or production scales compared to the known input during manufacturing.
 - 1.3. Inherent variance in manufacturing and testing (σ_1, σ_2 , etc.) can arise from multiple sources, including, but not limited to, raw material strength, batch addition variance, within-batch variation, between-batch variation, product weight (or dosing) variation, ingredient interactions, and analytical or microbiological method variation. These variances may be treated independently to allow for a simplistic summation.
 - 1.3.1. % Relative Standard Deviation (RSD) is equivalent to σ .
 - 1.3.2. Manufacturing or other tolerances ($\pm Z\%$) must be divided by 3 ($Z/3$) to be equivalent to σ .
 - 1.4. To enable sufficient process reliability, a confidence interval (CI) of 2 – 3 (95 – 99.7% confidence) is applied to the summation of variance.

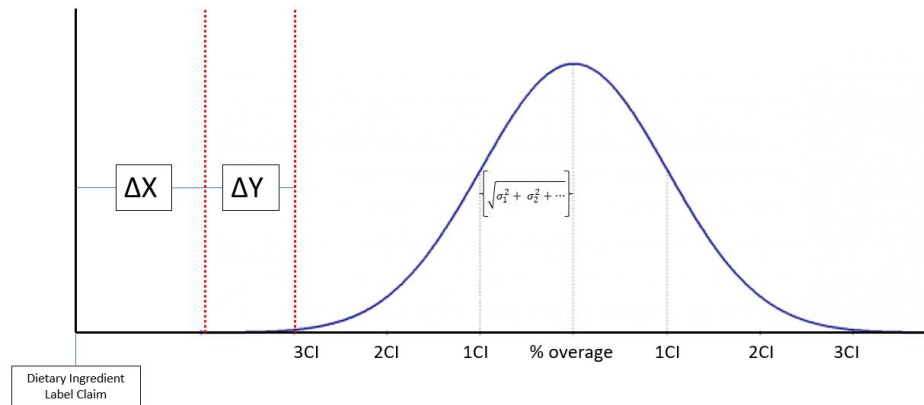


Standard Operating Procedure

OH-234 Determination of Dietary Ingredient Overages for Supplements and Natural Health Products

1.5. The mathematical expression described below may be leveraged to determine an overage.

$$\% \text{ overage} = \Delta X + \Delta Y + CI \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \dots}$$



- 1.6. In many cases, multiple sources of variation may not be known. However, dietary ingredient overages should be determined using anticipated losses (ΔX , ΔY), available sources of variation (σ), and appropriate confidence intervals (CI).
- 1.7. Observed dietary ingredient losses and variance from similar products, manufacturing processes, and/or models may be used to determine variable inputs (ΔX , ΔY , σ) to the equation provided in 1.5.
2. Once a decision regarding the overage of a dietary ingredient within a product formulation is made, complete Attachment 1 to document the decision and scientific rationale. This attachment must be filed with the Formulation Part (FOP) and approved by Quality Assurance.
3. Following assignment, the overage is documented in the master manufacturing record. Additionally, consistent with *OH-229 Product Specifications for Dietary Supplement and Natural Health Products* requirements, an upper limit specification is established to ensure the dietary ingredient remains in a specified range in the finished product. Account for inherent sources of variation identified in the overage calculation (σ_1 , σ_2 , etc.) in addition to raw material strength upper limit specifications when establishing finished product upper limit specification
4. The product formulation, which includes the overage, is evaluated by Product Safety and Regulatory Affairs during development and is documented in GPS clearance processes.
5. Considerations for Contract Manufacturers
 - 5.1. Contract manufacturers developing products for P&G are permitted to assign dietary ingredient overages for supplement and natural health products if determined to have the scientific expertise to do so. However, overage assignments will be reviewed and documented following the process outlined above prior to implementation.



Standard Operating Procedure

OH-234 Determination of Dietary Ingredient Overages for Supplements and Natural Health Products

REFERENCES

-	21 CFR 101, Food Labeling
-	21 CFR 111, Current cGMP in Manufacturing, Packaging, Labeling, or Holding Operations for Dietary Supplements
-	Quality of Natural Health Products Guide (Version 3.1, Date: May 1, 2015)
-	USP <2750> Manufacturing Practices for Dietary Supplements
-	Yoo, Seong Jae, et al. "Factors to consider in setting adequate overages of vitamins and minerals in dietary supplements." <i>Pharm. Forum.</i> Vol. 42. 2016
-	Cammarn, Stephen R. "Managing Overages in Vitamins, Minerals & Dietary Supplements." <i>CRN ASSOCIATE MEMBER SPOTLIGHT: EAS Consulting</i> , 30 Oct. 2020, www.crnusa.org/CRN-Daily-Supplement/Spotlight-EAS-Consulting
VV-QUAL-569947	OH-221 Management of Health Care Technical Standards in Enovia DSM
VV-QUAL-569949	OH-222 Determination of Nutrition of Foods and Supplements
VV-QUAL-1592906	OH-229 Product Specifications for Dietary Supplement and Natural Health Products
VV-QUAL-570035	OH-803 Pre-Market Stability Programs for Global Health Care Drug, Cosmetic, Medical Device, Dietary Supplement, Food, and Consumer Goods Products
VV-QUAL-570064	OH-825 Analytical Method Verification and Validation for Identity and Assay of Dietary Ingredients in Dietary Supplements and for Finished Product Dietary Supplements

SUPPORTING DOCUMENTS

VV-QUAL-2019451	Basis for Dietary Ingredient Overage
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SOP TRANSLATIONS

N/A	N/A	N/A
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DEFINITIONS & ABBREVIATIONS

P&F	Process & Formulation
MPD	Material, Process, Delivery
GPS	Global Product Stewardship
EBP	External Business Partner
DSHEA	Dietary Supplement Health and Education Act of 1994.
Dietary Supplement	US product classification for products intended to supplement the diet and contains dietary ingredient(s).
Natural Health Product (Canada)	A substance set out in Schedule 1 or a combination of substances in which all the medicinal ingredients are substances set out in Schedule 1, a homeopathic medicine, or a traditional medicine, that is manufactured, sold or represented for use in (a) the diagnosis, treatment, mitigation or prevention of a disease, disorder or abnormal physical state or its symptoms in humans. (b) restoring or correcting organic functions in humans; or (c) modifying organic functions in humans.
USP	United States Pharmacopeia
QNHPG	Quality of Natural Health Products Guide
Dietary Ingredient	Examples include Vitamins, Minerals, Amino Acids, Herbs, Botanicals, Enzymes, and those ingredients that do not have an established daily value as defined under 21 CCFR 101.36(b)(3).

Next Periodic Review Date: 31 Aug 2027

Last Periodic Review Date: Last Periodic Review Decision:

Page 5 of 6

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Standard Operating Procedure

OH-234 Determination of Dietary Ingredient Overages for Supplements and Natural Health Products

Class I Nutrient	Added nutrients in fortified or fabricated foods. 21 CFR 101.9, <i>Food Labeling</i>
Class II Nutrient	Naturally occurring (indigenous) nutrients. 21 CFR 101.9, <i>Food Labeling</i>
ΔX	Expected loss of dietary ingredient throughout expiry (expressed in % of label claim)
ΔY	Expected loss of dietary ingredients during processing $\Delta Y = \text{observed loss} / \text{initial quantity}$
σ	Any variance that may cause the dietary ingredient value to deviate from the intended formulated target (expressed in %). σ is calculated by standard deviation of the dataset divided by the average, then expressed as percentage.
Z	The one-sided tolerance for an allowable percentage range of values.
CI	Confidence Interval - A unitless, statistical measure which represents a probability range for observing an event within a given set of data
Label Claim	The quantity declared in the supplement facts panel for a dietary ingredient. For overage determination purposes: - Vitamins, Minerals, Amino Acids, Herbs, & Botanicals are overaged against the claimed quantity (i.e., milligram, gram) - If claimed, probiotics are overaged against the claimed colony forming unit (i.e., CFU) If claimed, enzymes are overaged against the claimed enzymatic activity

VERSION HISTORY (Last 3 Versions)

Version	Version Description
2	- Updated Step 3 to include further instruction on the establishment of upper limits for strength specifications - Included Step 5 – Considerations for Contract Manufacturers - Update to new SOP template
1	New

Document Approvals
Approved Date: 01 Aug 2025

Owner's Approval Task Verdict: Approve	Brian Laster, (laster.bj@pg.com) Owner 08-Jul-2025 15:18:16 GMT+0000
Approval Task Verdict: Approve	Kenneth Edelman, (edelman.kj@pg.com) Technical Approval 08-Jul-2025 15:36:51 GMT+0000
Approval Task Verdict: Approve	Trevor Kramer, (kramer.tm.1@pg.com) Technical Approval 09-Jul-2025 22:10:49 GMT+0000
Final Approver(s) Approval Task Verdict: Approve	Florence Iselin, (iselin.f@pg.com) Manager Approval 01-Aug-2025 14:49:57 GMT+0000

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